

Calibrated Selective Identity Association for Open-World Person Re-ID: A Three-Way MATCH/DEFER/UNKNOWN Decision Framework

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Abstract

Person re-identification systems typically force a decision on every query by accepting the top-1 nearest-neighbor match, even though a single similarity score does not distinguish a confident match from an ambiguous one or from a case where the true identity is simply absent from the gallery. We investigate a three-way alternative in which a lightweight, calibrated reliability estimator, placed on top of an unmodified detection, tracking, and retrieval pipeline, classifies each query as MATCH, DEFER, or UNKNOWN. The estimator combines similarity margin, an entropy-based measure of gallery ambiguity, crop quality, and, where video tracks are available, track-level temporal consistency, into a single probability using logistic regression. We evaluate this approach on two structurally different datasets: a large static-crop benchmark and a smaller real multi-camera video-track benchmark. Consistent with prior findings in open-set recognition, raw similarity is poorly calibrated, while margin alone substantially improves accept or reject decisions. Track-level temporal consistency, measurable only on video-track data, is the strongest additional signal identified beyond margin, and the combined model raises accuracy from 86.94 to 99.45 percent on a small (10-identity) video-track case study and from 95.34 to 99.12 percent on the static-crop dataset while still resolving a majority of queries automatically. Using a leave-one-identity-out open-set protocol, we find that DEFER and UNKNOWN are empirically separable, non-degenerate categories on both datasets, but that open-world rejection fails in two structurally different ways: a concentrated failure driven by specific look-alike identities, observed in a small 10-identity case study, and a diffuse failure that is consistent with gallery scale, observed on the larger dataset. Because the video benchmark has only ten identities, we treat this contrast as a hypothesis-generating observation rather than an established result. We report the absence of a single fully satisfactory UNKNOWN threshold on either dataset as a genuine finding, with direct implications for how open-set Re-ID rejection should be evaluated and deployed at different gallery scales.

1. Introduction

Person re-identification (Re-ID) systems typically retrieve the most similar gallery identity for a query image and accept or reject that match using a single fixed similarity threshold. This approach has two well-known weaknesses. First, similarity is not the same as confidence. A similarity score of 0.72 can reflect a clean, reliable match or a highly ambiguous one, depending on factors the raw score does not capture, such as the margin to the second-best candidate, the quality of the crop, and how consistent the match has been over time. A fixed threshold cannot distinguish these cases. Second, a binary accept/reject decision conflates two different failure modes. “This match is too uncertain to trust yet, but more evidence might resolve it” and “this person is genuinely not in the gallery” call for different downstream actions, yet a two-way decision treats them the same way.

Both problems are recognized in the broader open-set recognition (OSR) literature, where reject-option classifiers and calibration methods are well established [2], and the specific case of Re-ID has been posed in this way since at least the DIR/FAR framing in 2014 [4]. What remains unestablished is whether separating “uncertain” from “unknown” as genuinely distinct decisions, using signals available in a tracking pipeline rather than a single static image, measurably improves identity association, and whether such a three-way split is empirically real rather than a relabeling of standard rejection.

We investigate this by adding a lightweight, calibrated reliability estimator on top of an otherwise unmodified Re-ID pipeline (YOLO detection, a tracker, OSNet embeddings, and FAISS retrieval). Instead of a binary match/no-match output, the estimator produces one of three verdicts for each query: MATCH, indicating confidence sufficient to associate the query with a gallery identity; DEFER, indicating the match is too ambiguous to decide with current evidence; and UNKNOWN, indicating the evidence suggests the person is not present in the gallery. The estimator combines several signals, including similarity margin, gallery ambiguity (an entropy-based measure of how competitive the top candidates are), crop quality, and, where video tracks are available, track-level temporal consistency, into a single calibrated probability using logistic regression.

Margin-based rejection and general OSR calibration are established techniques and are not claimed as novel here. The two components investigated as new contributions are track-level temporal consistency as a reliability signal, and the explicit separation of DEFER from UNKNOWN as an empirically distinct decision rather than a notional one.

This paper is organized around three research questions. RQ1 asks whether raw Re-ID similarity is poorly calibrated as a standalone confidence measure. We expect this to hold, consistent with known OSR findings, but confirm it directly within this pipeline rather than assuming it transfers. RQ2, the first of two central questions, asks whether track-level temporal consistency improves calibration beyond what single-frame margin and quality features already provide. RQ3, the second central question, asks whether DEFER and UNKNOWN are empirically separable, that is, whether they behave as genuinely distinct categories of uncertainty or collapse into standard binary rejection under scrutiny. RQ3 is, in effect, the central test of this paper’s contribution: if DEFER turns out to be an empty or degenerate category, the added complexity of a three-way split is not justified.

We evaluate this approach on two structurally different benchmarks. Market-1501 provides a large gallery of approximately 750 identities with static, single-shot crops, and is used here for closed-set calibration and for synthetic open-set evaluation through identity withholding. PETS09 provides real multi-camera video with genuine tracklets, but at a very small scale of 10 identities, so we treat it as a case study rather than a full benchmark. It allows track-level temporal consistency to be computed and evaluated directly rather than only on static images. DukeMTMC was deliberately excluded from this work following its withdrawal over subject-consent issues. This pairing of datasets allows us to test whether our findings hold only for one particular data regime or generalize across both gallery scale and data modality.

Our results confirm that similarity margin remains the single most informative signal for match reliability, consistent with prior OSR findings (RQ1). Beyond margin, track-level temporal consistency (RQ2) is the strongest additional signal identified in this work, providing a larger and more consistent improvement over margin alone than crop quality or gallery ambiguity, though it is only measurable where genuine tracks exist, that is, on PETS09. For RQ3, we find that DEFER is a genuine, non-degenerate category on both datasets, containing a real mixture of correctly matched and genuinely absent queries at every uncertainty threshold tested, rather than collapsing to near-empty or near-total on either side. We also find that open-world rejection fails in two qualitatively different ways depending on dataset structure: a concentrated failure mode driven by specific look-alike identity pairs, observed in the small (10-

identity) PETS09 case study, and a diffuse failure mode that is consistent with gallery scale, observed on Market-1501. We report cases where our approach does not fully resolve, such as the absence of a single satisfactory UNKNOWN threshold on either dataset, as genuine findings rather than presenting only positive results.

This paper makes three contributions. First, we confirm that raw similarity is a poorly calibrated confidence measure in this pipeline and provide empirical evidence that track-level temporal consistency, a signal specific to tracking pipelines rather than static Re-ID, is the strongest additional reliability signal beyond margin (RQ1, RQ2). Second, we conduct a leave-one-identity-out open-set evaluation across two structurally distinct datasets showing that DEFER and UNKNOWN are empirically separable, non-degenerate categories (RQ3), directly validating the three-way MATCH/DEFER/UNKNOWN framing rather than assuming it. Third, we identify and analyze the root causes of two distinct open-world rejection failure modes, concentrated look-alike confusion (seen in a 10-identity video case study) and diffuse gallery-scale confusion (seen on Market-1501). To our knowledge these have not been previously characterized in this way, though the concentrated mode rests on only two confusing identities in a small dataset.

2. Related Work

Open-set recognition and reject-option classification.

The idea that a classifier should be able to abstain rather than force a decision is well established outside Re-ID. Reject-option classifiers, dating back to Chow’s [2] foundational formulation, and calibration methods for open-set recognition allow a model to withhold a prediction when confidence is low, rather than assign every input to one of a fixed set of known classes. We build directly on this line of work rather than replacing it: our MATCH and UNKNOWN verdicts correspond to standard accept and reject decisions in this framework, and our calibration methodology, fitting a probabilistic model over similarity-derived features and evaluating it with expected calibration error [5] and Brier score [1], follows established practice rather than introducing a new calibration technique.

Margin-based rejection in Re-ID. Within Re-ID specifically, the idea that the gap between the top match and the next-best candidate carries more information than the raw similarity score alone is a recognized rejection strategy, closely related to the detection-and-identification-rate evaluation framework introduced for open-set person re-identification by Liao *et al.* [4]. Our own baseline comparisons confirm this directly: margin consistently and substantially outperforms both a fixed similarity threshold and a calibrated similarity score, the latter obtained through Platt scaling [6], across both datasets we test. We treat this

finding as a replication of an established result within our specific pipeline, not as a novel contribution, and use margin as the foundation that our additional signals are compared against throughout the paper.

Retrieval accuracy versus association quality. Hou *et al.* [3] identify a mismatch between how Re-ID models are typically evaluated, as an image retrieval problem, and how they are actually used in multi-target multi-camera tracking, where the relevant question is whether two tracklets should be associated as the same identity. Their key empirical finding, that improvements in retrieval-style re-identification accuracy produce diminishing improvements in downstream tracking association, motivates a related but distinct concern in our work. Where Hou *et al.* focus on the mismatch between retrieval metrics and association outcomes, we focus on a mismatch within the retrieval decision itself: a single similarity score cannot distinguish a confidently correct match from a confidently wrong one, or from a case too ambiguous to decide either way. Track-level temporal consistency, one of the two signals we investigate as a new contribution, is our attempt to bring track-level information, in their spirit, into the confidence estimate itself rather than only into the evaluation protocol.

Positioning of this work. To our knowledge, no existing work combines an explicit three-way MATCH, DEFER, UNKNOWN decision with a track-level temporal consistency signal evaluated on real multi-camera video tracklets. Our contribution is not a new Re-ID backbone, detector, or tracker. It is a lightweight reliability layer placed on top of an unmodified pipeline, together with an empirical test, using held-out identities on two structurally different datasets, of whether the three-way split is a genuine distinction or a relabeling of standard binary rejection.

3. Method

3.1. Datasets and PETS09 crop construction

PETS09 is distributed as multi-view, full-frame video rather than as a set of cropped, identity-labeled images, so it could not be used directly in the Market-1501 style. To build a crop-level benchmark, we ran a YOLO detection, BoT-SORT tracking, and OSNet embedding pipeline over the video to extract person crops into one folder per identity, and we recorded the crop-to-identity assignments in a JSON annotation file. Because this automatic pipeline is imperfect, its output contained identity errors of two kinds: a single real person being assigned several different track identities (identity fragmentation), and one track identity being assigned to more than one real person (identity merging). We corrected these errors manually, so the final identity labels reflect manual verification rather than raw tracker out-

put, and the final PETS09 benchmark contains 4,501 crops after correction. We note two consequences. First, the same detection and tracking components used to build the crops (YOLO and BoT-SORT) are also the ones used in the evaluated pipeline; only the OSNet embeddings were recomputed on the corrected crops. This may make the crops more favorable to the evaluated pipeline than independently annotated crops would be. Second, the manual correction was performed by a single annotator, so we did not measure annotation agreement. This is a further reason to treat PETS09 as a small case study rather than a full benchmark.

3.2. Pipeline

Our system builds on an unmodified Re-ID pipeline. Detection uses YOLO26x. Tracking uses BoT-SORT. Appearance embeddings come from OSNet (osnet_x1_0), using a checkpoint pretrained on Market-1501 (osnet_x1_0_market1501.pth), producing 512-dimensional embeddings. Retrieval uses a FAISS IndexFlatIP index over L2-normalized embeddings, so that inner product is equivalent to cosine similarity. We make no changes to any of these components. Instead, we add a lightweight reliability estimator that sits after retrieval and converts a raw similarity score into a calibrated probability, together with a three-way decision rule that maps that probability onto a MATCH, DEFER, or UNKNOWN verdict.

We note that because the OSNet checkpoint used for embeddings was itself pretrained on Market-1501, the Market-1501 evaluation in this paper is not a fully held-out test of the embedding backbone: the backbone was trained on data from the same identity distribution it is evaluated on. This does not affect the PETS09 evaluation, which uses a structurally different dataset than the one the backbone was pretrained on, but it should be kept in mind when interpreting the absolute accuracy figures reported for Market-1501.

3.3. Signals

The reliability estimator combines up to four features, computed per query.

Similarity margin is the difference between the top-1 similarity score and the highest similarity score among gallery candidates with a different identity than the top-1 pick. This distinction matters because a naive margin, computed as the gap between the top-1 and literal rank-2 score regardless of identity, is frequently measuring redundancy within the same identity’s gallery crops rather than genuine ambiguity between different people, since most Re-ID galleries contain multiple images per identity. Using the nearest score from a genuinely different identity corrects for this.

Gallery ambiguity is an entropy-based measure of how competitive the top candidates are. We take the top- k similarity scores among different-identity candidates ($k = 2$), rescale them by a temperature parameter, and apply a softmax to obtain a probability distribution over candidates, then compute the entropy of that distribution. Low entropy indicates one candidate clearly dominates. High entropy indicates two or more candidates are close enough that the ranking is effectively undecided. The temperature parameter must be tuned separately for each dataset, since raw similarity gaps between different identities can vary substantially in scale across datasets. We verified this concretely: on Market-1501, a temperature of 0.005 gave a well-behaved entropy distribution, but reusing the same value on PETS09 produced entropy values pinned near the theoretical maximum for a large fraction of queries, since PETS09’s real top-2 different-identity similarity gaps are considerably smaller. We selected a dataset-specific temperature of 0.002 for PETS09, and 0.005 for Market-1501, by sweeping a range of values and checking the resulting entropy distribution directly against ground-truth correctness. For both datasets, this sweep used the full query set prior to the fit/evaluation split described in Section 3.4, so the evaluation half was not fully held out from this one hyperparameter choice. The effect is likely small, since the criterion was whether entropy values were pinned near the maximum rather than maximizing accuracy, but we note it as a limitation (see Section 5.8).

Crop quality is measured as the variance of the Laplacian of the grayscale image, a standard measure of image sharpness, computed for both the query crop and the matched gallery crop, with the lower of the two values used as a single combined feature. We also computed resolution and aspect-ratio deviation as candidate quality features, but found these carried no signal on Market-1501, where all images are stored at a fixed resolution prior to any processing, and we accordingly retained blur as the only informative quality feature.

Track-level temporal consistency is defined only where a query corresponds to a sequence of frames from the same tracked individual, as in PETS09’s video data, rather than a single static image, as in Market-1501. For each track, we compute the top-1 identity prediction independently for every frame, take a majority vote across the track, and compute the fraction of frames whose individual prediction agrees with that majority. This fraction is then assigned to every frame in the track as a per-frame feature. The intuition is that a track whose frame-level predictions agree with each other consistently is more trustworthy than one whose predictions vary from frame to frame, independent of any single frame’s own similarity score.

3.4. Combining signals

For each dataset, we fit a logistic regression model (scikit-learn’s `LogisticRegression` with default settings) over the available signals, using standardized features, to produce a single calibrated match probability per query. Market-1501 combines margin, crop quality, and gallery ambiguity, since no track-level information is available for static single-shot crops. PETS09 combines margin, gallery ambiguity, and track-level temporal consistency. We initially included crop quality as a fourth PETS09 signal, but found its fitted coefficient was negligible and that removing it left the model’s coverage-accuracy performance essentially unchanged, so we report PETS09’s three-signal model, excluding crop quality, as the final combined model for that dataset.

To avoid circularity between fitting and evaluation, we split each dataset’s query set in half at the query level, stratified by whether the top-1 match was correct, using one half to fit the logistic regression and reserving the other half exclusively for evaluation. All reported coverage, accuracy, and calibration results are computed on the held-out evaluation half only. On PETS09, where queries are individual frames drawn from continuous tracks, this means frames from the same track can appear in both halves. Because the track consistency feature is computed per track, this may make the PETS09 fit and evaluation halves less independent than a track-level split would, and could make our PETS09 results somewhat optimistic. We report this as a limitation (see Section 5.8).

3.5. Thresholding

The fitted model’s output probability is converted into a MATCH, DEFER, or UNKNOWN verdict using two thresholds. Above the MATCH threshold, a query is accepted as a confident match. Below the UNKNOWN threshold, a query is treated as evidence that the true identity is not present in the gallery. Everything between the two thresholds is deferred.

We set the MATCH threshold by sweeping a range of candidate probability cutoffs on the evaluation set and selecting the lowest cutoff at which accuracy reaches a target of 99 percent, since beyond this point further increases in the cutoff produce only small additional gains in accuracy at a disproportionate cost in coverage. This method was applied identically to both datasets.

Setting the UNKNOWN threshold requires knowing how the model behaves on queries whose true identity is genuinely absent from the gallery, which does not occur naturally in either dataset’s standard evaluation protocol. We constructed such queries using a leave-one-identity-out procedure. For a sample of identities, we removed that identity’s crops entirely from the gallery, then evaluated all of that identity’s query frames against the resulting, now gen-

uinely identity-free, gallery. On PETS09, we performed this procedure for all 10 identities. On Market-1501, given the substantially larger identity pool, we sampled 60 of approximately 750 identities. In both cases, every resulting query is guaranteed to produce an incorrect top-1 match, since the true identity is no longer retrievable, allowing us to observe the model’s output probability under conditions of genuine absence.

We evaluated candidate UNKNOWN thresholds using a containment framework, in which a target containment level specifies what fraction of genuinely absent queries should fall below the threshold, and we measured the resulting cost, defined as the fraction of genuinely correct matches in the standard evaluation set that would be incorrectly classified as UNKNOWN at that same threshold. We report results across four containment levels, 80, 85, 90, and 95 percent, for both datasets.

4. Results

4.1. RQ1: Is raw similarity poorly calibrated?

Across both datasets, raw similarity alone was a substantially weaker basis for accept or reject decisions than similarity margin. Because the OSNet embedding backbone used throughout this work was pretrained on Market-1501 (Section 3.2), the Market-1501 numbers reported below, including the baseline comparison in Table 1, reflect within-distribution performance on identities the backbone may have already seen during pretraining, rather than a test of generalization to unseen identities; PETS09 is not affected by this, since it is structurally distinct from the backbone’s training data. Table 1 compares Market-1501’s four baseline methods at matched coverage. A fixed similarity threshold gave only a modest coverage-accuracy tradeoff, and Platt-scaled calibration of the same similarity score improved on it only slightly, indicating that rescaling similarity into a probability does not by itself add new information. Margin clearly and consistently outperformed both raw and calibrated similarity, and we confirmed this improvement was genuine, not an artifact of evaluating on a smaller sample, by comparing margin-based rejection against random rejection at matched coverage. These results confirm that a single similarity score is a poorly calibrated confidence measure, consistent with prior findings in open-set recognition, and motivate the use of margin, together with additional signals, as the basis for our combined reliability estimator.

Table 2 makes this improvement concrete at the level of the full system. Forcing a decision on every query, equivalent to always accepting the top-1 match, gives 86.94 percent accuracy on the PETS09 case study and 95.34 percent accuracy on Market-1501. Allowing the combined model to select only its most confident predictions raises accuracy to

Table 1. Market-1501 baseline comparison at matched coverage (90.6% coverage shown as representative).

Method	Accuracy
Fixed similarity threshold	97.09%
Platt-scaled calibrated similarity	97.23%
Corrected margin	98.45%

Table 2. Combined model versus raw top-1 matching, no rejection versus selective MATCH threshold.

Dataset	Raw top-1 acc.	Combined acc. at thresh.	Coverage retained
PETS09 (10-id. case study)	86.94%	99.45%	59.0%
Market-1501	95.34%	99.12%	83.0%

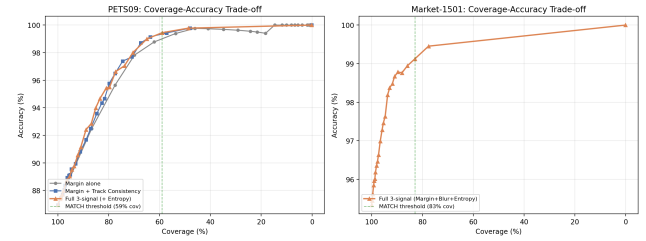


Figure 1. Coverage-accuracy trade-off for PETS09 (left) and Market-1501 (right).

99.45 percent on PETS09, while still retaining 59.0 percent of all queries as confident matches, and to 99.12 percent on Market-1501, while retaining 83.0 percent of all queries. This is the practical value of selective prediction: a large majority of queries can still be resolved automatically and confidently, while the harder minority is correctly identified as needing further evidence rather than being silently mismatched.

4.2. RQ2: Does track-level temporal consistency improve calibration?

Figure 1 shows full coverage-accuracy curves for both datasets. On this small (10-identity) PETS09 video benchmark, adding track-level temporal consistency to margin improved accuracy at every coverage level tested, with no exceptions, typically by 0.3 to 1.0 percentage points in the practically useful middle range of coverage. This is a considerably larger and more consistent gain than crop quality or gallery ambiguity produced on Market-1501, where the equivalent combined model improved on margin alone by only 0.1 to 0.2 percentage points, and did so inconsistently across the coverage range.

Table 3 summarizes each signal’s direct separation between correctly and incorrectly matched queries, shown visually in Figure 2. Track-level temporal consistency

Table 3. Signal separation between correct and incorrect matches.

Signal	Dataset	Corr.	Incorr.	Corr. w/ marg	Metric	PETS09	Market-1501
Track consistency	PETS09	0.892	0.719	0.53	Expected calibration error	0.0278	0.0111
Gallery ambig. (ent.)	PETS09	0.232	0.324	−0.06	Brier score	0.0787	0.0319
Crop quality (query blur)	M-1501	1094	955	n/m	Top bin (0.9–1.0) share of data	65%	87.6%
Crop quality (gallery blur)	M-1501	1147	989	n/m	Top bin calibration gap	0.0103	0.0025

n/m = not measured.

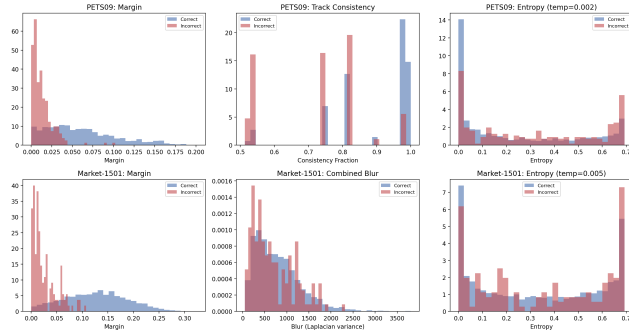


Figure 2. Per-signal distributions for correct vs. incorrect matches, PETS09 and Market-1501.

Table 4. Fitted logistic regression coefficients, standardized features.

Signal	PETS09	Market-1501
Margin	2.38	3.20
Track consistency	0.70	n/a
Gallery ambiguity (entropy)	−0.31	−0.40
Crop quality (blur)	excluded	−0.20

n/a = not applicable. “Excluded” indicates the fitted coefficient was negligible and the signal was dropped from PETS09’s final model (see Section 3.4).

showed the clearest separation of any secondary signal identified in this work, and was only moderately correlated with margin, indicating it captures related but non-redundant information.

Margin remained the dominant signal in every fitted model, as shown in Table 4. We do not interpret our results as showing that track consistency exceeds margin in importance. Rather, among the signals evaluated beyond margin itself, track-level temporal consistency provided the largest and most consistent additional improvement, supporting its use as a genuinely informative complementary signal specific to tracking pipelines rather than static image retrieval.

Crop quality’s fitted coefficient on PETS09 was negligible, and removing it from the combined model left coverage-accuracy performance essentially unchanged. We report PETS09’s final combined model using only margin, gallery ambiguity, and track consistency, and treat this as a genuine finding: crop quality’s usefulness as a reliability

Table 5. Calibration metrics, evaluation set.

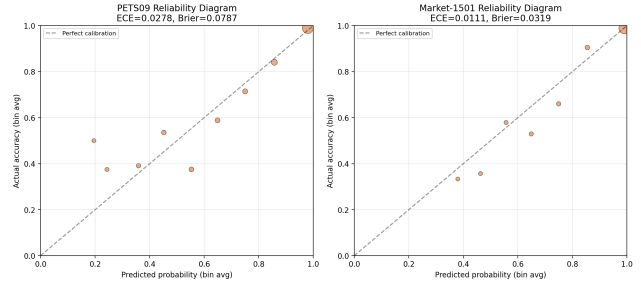


Figure 3. Reliability diagrams for PETS09 and Market-1501 combined models.

Table 6. MATCH thresholds.

Dataset	Threshold	Accuracy	Coverage
PETS09	0.9328	99.45%	59.0%
Market-1501	0.9476	99.12%	83.0%

signal does not transfer consistently across the two dataset regimes tested here.

Table 5 and Figure 3 summarize calibration for both combined models. Both are well calibrated overall, with the strongest calibration in the high-confidence region where the majority of predictions fall.

4.3. RQ3: Are DEFER and UNKNOWN empirically separable?

We first set a MATCH threshold on each dataset by identifying the lowest probability cutoff at which evaluation-set accuracy reached 99 percent, shown in Table 6.

Setting a corresponding UNKNOWN threshold proved considerably harder. Figure 4 shows the distribution of model probabilities for closed-set queries against open-set queries constructed by removing an identity from the gallery entirely. Table 7 shows the cost, in wrongly rejected genuine matches, of catching increasing proportions of genuinely absent queries. No threshold on the 10-identity PETS09 benchmark achieved an acceptable trade-off. Market-1501 showed a more favorable, though still imperfect, tradeoff, with costs roughly half those of PETS09 at every level tested.

We investigated the source of PETS09’s poor separability directly, rather than treating it only as a limitation of

Table 7. UNKNOWN threshold containment sweep: cost of catching genuinely absent queries, in percent of genuine matches wrongly rejected.

Containment target	PETS09	Market-1501
80%	18.93%	9.69%
85%	21.11%	10.78%
90%	25.46%	12.42%
95%	30.90%	16.54%

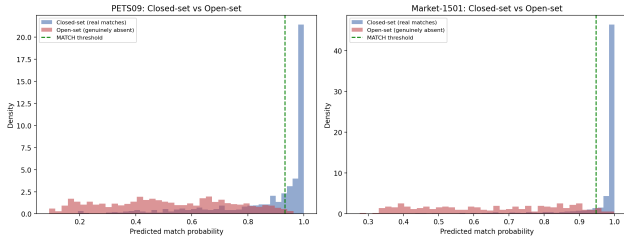


Figure 4. Closed-set vs. open-set predicted match probability distributions, PETS09 and Market-1501.

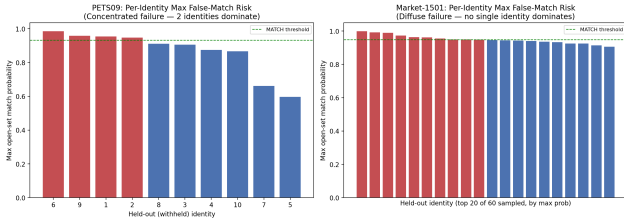


Figure 5. Per-identity maximum false-match risk under leave-one-identity-out evaluation, PETS09 and Market-1501.

the model. Figure 5 shows the maximum open-set match probability for each withheld identity on both datasets. On this small (10-identity) PETS09 benchmark, two identities out of ten accounted for the overwhelming majority of confident false matches, and in both cases the system consistently misidentified the withheld individual as the same specific gallery identity, in 96.7 and 89 percent of frames respectively. With only ten identities, this is a pattern from two cases, not a statistical estimate. Both identities also had noticeably blurrier query crops than the dataset average. On Market-1501, the equivalent false matches were spread across ten different gallery identities among the sixty sampled, with no repeated confusion pattern. We interpret this as evidence of two structurally different open-world failure modes: a concentrated failure driven by specific visually similar identity pairs, compounded by lower image quality, observed in the small 10-identity PETS09 case study, and a diffuse failure that is consistent with gallery scale itself, observed on the large-gallery dataset.

Given this, rather than commit to a single UNKNOWN threshold on either dataset, we examined how the DEFER

Table 8. DEFER composition across containment targets.

Containment	PETS09		Market-1501	
	real match	absent	real match	absent
80%	36.8%	63.2%	73.3%	26.7%
85%	41.6%	58.4%	77.4%	22.6%
90%	46.4%	53.6%	82.6%	17.4%
95%	56.0%	44.0%	80.0%*	20.0%*

*Market-1501’s 95% row is based on only 10 total DEFER items and should be treated as unreliable due to sample size.

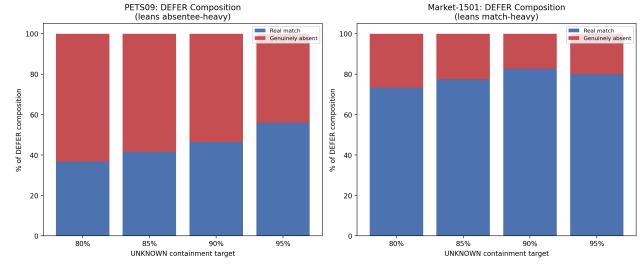


Figure 6. DEFER composition across containment targets, PETS09 (absentee-heavy) and Market-1501 (match-heavy).

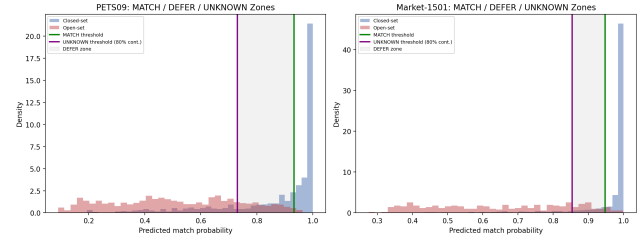


Figure 7. MATCH/DEFER/UNKNOWN zones with thresholds marked, PETS09 and Market-1501.

category behaves across the full range of containment targets tested. Table 8 and Figure 6 show DEFER’s composition, in terms of the proportion of genuinely correct matches versus genuinely absent queries it contains, at each containment level.

On both datasets and at every containment level tested, DEFER contained a real, non-degenerate mixture of both categories rather than collapsing to near-total or near-empty on either side. The direction of this mixture differed consistently between datasets: PETS09’s DEFER category leaned toward containing genuinely absent queries, consistent with its concentrated open-set failure, while Market-1501’s DEFER category leaned toward containing genuinely correct matches, consistent with its MATCH threshold covering a smaller proportion of its overall base rate. Figure 7 shows both distributions with the MATCH and UNKNOWN thresholds and the resulting DEFER zone marked directly.

Taken together, these results support a positive answer

to RQ3. DEFER and UNKNOWN are empirically separable and behave as genuinely distinct categories rather than an arbitrary relabeling of standard rejection, and the composition of DEFER reflects, in an interpretable way, the underlying structure of each dataset’s open-world uncertainty, rather than being a fixed or arbitrary property of the model itself.

5. Discussion

5.1. What the combined model actually buys

The central practical result of this work is that selective prediction meaningfully outperforms forced matching (Table 2). A lightweight reliability layer, added without modifying the underlying Re-ID pipeline, is enough to substantially reduce the practical cost of identity errors while still resolving a majority of queries automatically. This result does not require the three-way DEFER and UNKNOWN split to hold; it would remain true even under a simpler two-way accept or reject framing. The value of the three-way split specifically rests on RQ3, discussed further below.

5.2. Track consistency as a genuinely new signal, with a caveat

Our RQ2 results show that track-level temporal consistency is the strongest additional signal we identified beyond margin, with a clearer separation between correct and incorrect matches than any other secondary signal tested, and only moderate correlation with margin itself. This supports the proposal’s central hypothesis that information specific to tracking, rather than to any single frame, carries genuine value for reliability estimation. The caveat is that this signal is, by construction, unavailable for static single-shot Re-ID, and its value could only be demonstrated on the 10-identity PETS09 case study, the smaller and less standard of our two datasets. Whether track consistency continues to provide a comparable improvement on longer, noisier, or more crowded video sequences than PETS09 provides remains untested here.

5.3. Why crop quality did not transfer

Crop quality, measured as image blur, showed a small but genuine effect on Market-1501 and no measurable effect on PETS09, to the point that removing it from PETS09’s combined model changed performance negligibly. We do not believe this reflects an error in the feature itself. A more likely explanation is that Market-1501’s crops vary more in isolation, since they are drawn from many separate capture instances, while PETS09’s crops come from continuous video tracks in which blur levels are comparatively stable within a track and only vary meaningfully between tracks, a distinction the per-frame margin and entropy signals may already capture indirectly through the resulting

similarity scores. We report this as a genuine negative finding rather than omit it, since it suggests that crop-quality features tuned on static Re-ID benchmarks should not be assumed to transfer to video-based deployment without direct validation.

5.4. The open-world rejection problem does not have one shape

Perhaps the most significant finding of this work is that open-set rejection failed for two different reasons on our two datasets, and that these reasons have different practical implications. In the 10-identity PETS09 case study, failure was concentrated: two identities out of ten were confidently and repeatedly mistaken for one specific gallery identity. Because this rests on two identities in a very small gallery, we present it as a pattern worth testing on larger video datasets, not as a general property of video Re-ID. This kind of failure is, in principle, addressable, either by improving crop quality for the affected cases or by adding a signal that measures absolute resemblance to a specific candidate rather than only the relative ranking among candidates, which is what all four of our current signals measure. On Market-1501, failure was diffuse, consistent with the explanation that a larger gallery mechanically increases the chance of an incidental high-similarity match somewhere in the pool. This kind of failure is harder to address through a single fix, since it is a structural consequence of gallery size rather than a property of any specific identity pair, and it implies that open-set false-match rates should be expected to change, most plausibly to worsen, as a deployed system’s gallery grows, independent of any change in the underlying Re-ID model.

5.5. Why we did not force a single UNKNOWN threshold

We consider the absence of a clean UNKNOWN threshold on either dataset a genuine result rather than an unresolved gap. Table 7 shows every containment level carried a real cost, and on PETS09 a severe one. Reporting a single chosen threshold without this context would overstate the reliability of the UNKNOWN category and obscure a limitation that matters for deployment, since a system operator would need to accept a specific, dataset-dependent trade-off between catching absent individuals and preserving correct matches, rather than assume a threshold exists that does both well.

5.6. DEFER’s composition is informative, not fixed

Our RQ3 results show that DEFER is not merely a leftover middle zone between two arbitrary cutoffs, but a category whose composition tracks the underlying structure of each dataset’s uncertainty in an interpretable way, leaning toward genuinely absent queries on the dataset with concentrated

open-set failure and toward genuinely correct matches on the dataset with diffuse failure. We note, however, that this composition shifted as we varied the containment target, particularly on PETS09, where the share of absent queries in DEFER fell as the containment target rose (Table 8). This means DEFER’s composition should be understood as a property of where the UNKNOWN threshold is placed relative to a dataset’s uncertainty structure, not as a fixed characteristic of the dataset or model alone.

5.7. A note on the phrase “driven by gallery scale”

Throughout this paper we describe Market-1501’s diffuse failure as “consistent with” gallery scale rather than “driven by” it. Our data shows the failure is spread across ten different identities with no repeated confusion pattern, which is consistent with a gallery-size explanation, but we did not directly test the effect of varying gallery size (for example, by evaluating on random subsets of Market-1501’s identity pool), so we treat the gallery-scale account as a plausible explanation rather than a directly established causal claim.

5.8. Limitations

This work has several limitations beyond those already discussed. Our Market-1501 open-set evaluation sampled 60 of approximately 750 identities rather than testing every identity, for computational reasons, and while this sample is unlikely to be systematically biased, a full leave-one-identity-out evaluation could reveal a different or more extreme failure pattern than the diffuse pattern we observed. Our calibration results, while strong in aggregate, are based on comparatively few examples in the low-confidence probability range on both datasets, since the base rate of correct matches is high on both, and we do not consider calibration in this range well established by our results. Finally, both datasets used here are moderate in scale relative to real deployment settings, PETS09 in particular has only ten identities, and the generalization of our findings, especially the specific concentrated-versus-diffuse failure distinction, to substantially larger or more visually homogeneous populations remains an open question.

Two further limitations concern our evaluation protocol. First, the PETS09 fit/evaluation split (Section 3.4) was made at the query (frame) level rather than the track level, so frames from the same track can appear in both the half used to fit the logistic regression and the half used for evaluation. Because the track consistency feature is computed per track, this may make the PETS09 fit and evaluation halves less independent than a track-level split would produce, and could make our PETS09 results somewhat optimistic. Second, the entropy temperature parameter (Section 3.3) was tuned by sweeping values against the full query set for each dataset, prior to the fit/evaluation split, rather than on the fit half alone, so the evaluation half was not fully held out

from this hyperparameter choice; we believe the effect of this is likely small, since the tuning criterion was qualitative (whether entropy values were pinned near their maximum) rather than accuracy-maximizing, but we note it here for completeness.

Third, PETS09 identity labels were built by an automatic tracker and then manually corrected by a single annotator, which may leave residual label noise. Fourth, the OSNet embedding backbone used throughout this work was pre-trained on Market-1501, meaning the Market-1501 evaluation is not a fully held-out test of the embedding model itself; this does not apply to PETS09, which is a structurally distinct dataset from the backbone’s training data, but it should be considered when comparing absolute accuracy figures between the two datasets. This concern is sharper for the leave-one-identity-out open-set experiments (Section 3.5, Table 7, Figure 5) than for the closed-set baseline numbers alone: removing an identity’s crops from the gallery removes that identity from retrieval, but it does not undo whatever the embedding space already learned about that identity’s appearance during pretraining, so our Market-1501 open-set results may understate the difficulty a truly unseen identity would pose, in a way that does not apply to PETS09.

6. Conclusion

This paper examined whether Re-ID systems benefit from replacing forced nearest-neighbor matching with a three-way MATCH, DEFER, UNKNOWN decision, made by a lightweight, calibrated reliability estimator placed on top of an unmodified detection, tracking, and retrieval pipeline. Across two structurally different datasets, a large static-crop benchmark and a smaller real video-track benchmark, we found that raw similarity is a poorly calibrated basis for this decision, confirming prior findings in open-set recognition within our specific pipeline. We found that track-level temporal consistency, available only where genuine tracking data exists, is the strongest additional signal beyond similarity margin, supporting the use of track-level information for reliability estimation rather than confidence estimation based on single frames alone. Most centrally, we found that DEFER and UNKNOWN behave as genuinely separable categories on both datasets, rather than a relabeling of standard binary rejection, and that the specific way open-world matching fails differs meaningfully with gallery scale and structure: a concentrated failure driven by specific look-alike identities in a small 10-identity gallery, and a diffuse failure consistent with gallery size on a larger one. Because the small-gallery result comes from a single ten-identity video benchmark, confirming it on larger video datasets is the most important next step.

We reported these findings, including the cases where our approach did not fully resolve, such as the absence of a

single reliable UNKNOWN threshold on either dataset, as genuine results rather than limitations to be minimized. We believe this is the more useful contribution for practitioners considering selective prediction for open-world Re-ID: not a claim that uncertainty can always be cleanly separated into confident categories, but evidence for where it can, where it cannot, and why the two datasets tested here differed in this respect.

Future work could extend this evaluation to larger and more diverse video-track datasets to test whether the concentrated-versus-diffuse failure distinction identified here generalizes beyond the two datasets tested, and could investigate additional signals capable of measuring absolute resemblance to a specific candidate identity, rather than only relative ranking among candidates, which may be necessary to address the concentrated failure mode observed on the smaller dataset in this work.

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